

Multidimensional Real Analysis - Duistermaat, Kolk

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1. Chapter 1 - Continuity

(a) section 1.1 - Inner product and Norm

- (Defn) Inner product and Norm
- (Thm) Properties of Inner Product
- (Thm) Cauchy-Schwarz Inequality

Proof Idea: Decompose a vector x as its projection onto y plus its orthogonal component. Take the norm squared and solve for the orthogonal component. Its nonnegativity forces the CS inequality.

- (Thm) Inequalities related to the norm
 - Triangle inequality and extended triangle inequality
 - Reverse triangle inequality
 - $|x_j| \leq \|x\| \leq \sum_{i=1}^n |x_i| \leq \sqrt{n}\|x\|$

Proof Idea: First inequality is straight forward, second inequality can be seen by decomposing the vector and applying the triangle inequality. Third inequality can be seen by forming the vector x_{abs} composed of the absolute values. Then take the inner product with the vector of all 1s and apply C.S.

- $\max_j |x_j| \leq \|x\| \leq \sqrt{n} \max_j |x_j|$. This implies that for a cube of side length 2 is enclosed in a sphere of radius $\sqrt{n}$ which is enclosed in a cube of side length $2\sqrt{n}$.

Proof Idea: First inequality follows from the previous part. 2nd inequality can be seen by replacing each component of x with the maximum component. The norm of the new vector is the RHS.

- (Defn) Distance between two vectors
- (Defn) Convergence or limit of a sequence of vectors in $\mathbb{R}^n$.
- (Thm) A sequence of vectors converges iff each sequence of components converges.
- (Thm) The limit operation on vectors is still linear.

(b) Section 1.2 - Open and closed sets

- (Defn) Open Ball/neighborhood